

Normalized Cross-Correlation

- The brightness is misleading?
Remove the averages $\bar{\mathbf{w}}_{ij}$, $\bar{\mathbf{F}}$

$$\mathbf{w}_{ij} - \bar{\mathbf{w}}_{ij}$$

Patch / window within input image,
centered at $[i,j]$ adjusted by average

$$\mathbf{F} - \bar{\mathbf{F}}$$

Average-adjusted kernel

- Normalize by the l_2 norm $\|\cdot\|$

$$\hat{\mathbf{w}}_{ij} = \frac{\mathbf{w}_{ij} - \bar{\mathbf{w}}_{ij}}{\|\mathbf{w}_{ij} - \bar{\mathbf{w}}_{ij}\|}$$

$$\hat{\mathbf{F}} = \frac{\mathbf{F} - \bar{\mathbf{F}}}{\|\mathbf{F} - \bar{\mathbf{F}}\|}$$

l_2 norms: sqrt of the sum of the squared elements

- Compute the weighted sum

$$x_{ij} = \hat{\mathbf{w}}_{ij}^T \hat{\mathbf{F}} = \sum_{u,v} \hat{\mathbf{w}}_{ij} \hat{\mathbf{F}}$$

Sum the element-wise multiplication;
note the matrix notation alternative.